SYNthia: An Interface Concept for Writing With Large Language Models

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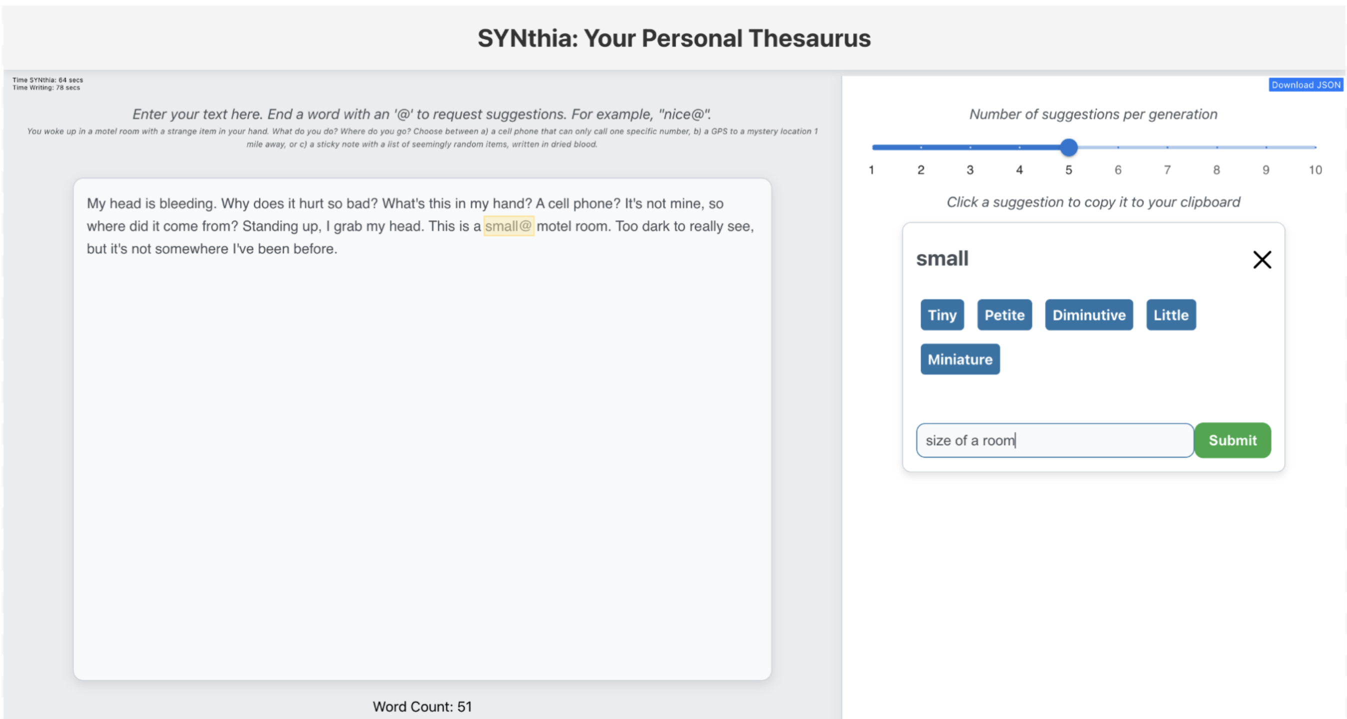

Figure 1: SYNthia's interface. On the left, users can create/edit their writing in the text box. SYNthia is prompted with the @ symbol (see highlighted word "small"), triggering the suggestion box to pop up on the right. Users can toggle the number of suggestions with the slider on the top right corner, as well as interact with the suggestion boxes by providing additional context for reprompts. Additional measurements are collected via a dual timer system (top left corner) and a live tracker of prompted words, context given, and word chosen through a JSON file (top right corner).

Abstract

Artificial Intelligence (AI) can assist writers, but may also generate distracting or imperfect suggestions. Word choice presents a challenge for writers that is often addressed via thesaurus usage, but online thesauruses can disrupt writing flow by requiring context switching, and generally lack context awareness. We present SYNthia, an LLM-based word suggestion interface that lets users request synonyms and provide natural language context. Although we hypothesized that novice and expert writers would use SYNthia differently, in a user study (n=19) we found no significant differences, raising more questions for further study. However, qualitative results revealed that adding context increased cognitive load, but also encouraged critical thinking about word choice and even plot. Even when suggestions were rejected, SYNthia's suggestions influenced writer decisions. These findings make progress towards the greater goal of integrating AI in the writing process while maintaining human agency and ownership. All code can be found at the Github repository: https://github.com/iliang1234/SYNthia

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CCS Concepts

• **Human-centered computing** → **User interface programming**; • **Computing methodologies** → *Knowledge representation and reasoning*; **Cognitive science**; • **Applied computing** → *Computer-managed instruction.*

Keywords

AI-assisted writing, Large language models (LLMs), Creative writing, Cognitive process, Thesaurus, Metacognition, Ownership, Co-creativity

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1 Introduction

In 2023, Stephen Marche published *Death of an Author*, a 95% AI-generated novel, the first of its kind reviewed by The New York Times. Some called it "groundbreaking" [24], while others a "tumor through the spine" of literature [1]. Marche, however, compared AI-assisted writing to hip-hop sampling, arguing that if artists like Ava Max can heavily sample music [17], writers should be able to do the same with AI [1]. While perhaps extreme, Marche's book demonstrates how artificial intelligence—in particular large language models (LLMs)—is becoming widely discussed in creative writing communities.

LLMs are increasingly used in creative writing, such as to aid in brainstorming and style refinement [16], but raise concerns about originality [2], authorship [13], and over-reliance [26]. In this work, we explore how LLM-generated word suggestions shape novice and expert writers' thought processes. To explore this question, we created SYNthia, an LLM-powered thesaurus where suggestions can be refined based on context explicitly added by the user.

We conducted a between-subjects user study with novice (n=12) and expert (n=7) writers, where participants wrote a short story for 15 minutes while making use of SYNthia. Participants requested word suggestions nine times on average, accepting suggestions 43% of the time. We saw no difference in interaction patterns between novice and expert writers. However, participants reported that adding context was energy-intensive, but also triggered critical thinking. Sometimes, even when suggestions were unsatisfactory, participants felt greater confidence in their original word choice.

Despite these interactions, 74% claimed there was no AI influence on their work because they viewed SYNthia as just a tool, with a small scope of influence and multiple manual components. Most participants found the tool inspiring and helped them shape the story's tone, direction, and structure.

In summary, we report on:

- Design of SYNthia, an LLM-powered thesaurus enabling in-context word suggestions.
- Quantitative evaluation that yielded no significant differences by writing expertise or AI expertise.
- Qualitative evaluation exploring AI's role in writing, system efficiency, and contextual influence.
- Implications of AI in creative writing.

2 Related Works

2.1 Cognitive Processes of Novices vs. Experts

Research has found that expert and novice writers differ in their cognitive processes, including how they plan and review their work [29]. Experts engage in more extensive planning before and during the writing process, considering audience and goals, while novices focus on task completion [5]. Experts also approach reviewing more holistically, revising structure and coherence, whereas novices tend to make surface-level edits [5]. Metacognition, or self-monitoring and self-reflecting, also plays a key role in the writing process. Expert writers often demonstrate stronger metacognitive skills by critically assessing their word choices [14] and strategizing improvements, while novices struggle with this awareness and rely more on external assistance, such as AI tools [4, 28]. We hypothesized that experts in our user study would be more critical of SYNthia's suggestions.

2.2 Computer-Assisted Writing

Writing tools, from spellcheckers to AI-driven text suggestions, continue to evolve [30, 32, 33]. In creative writing, AI enhances word choice, description, and world-building. Lee et al. [21] define a design space for AI writing assistants, emphasizing adaptability and interactivity—key to SYNthia's approach. Existing tools like Metaphoria [15] and Slogan Tools [10] ensure AI suggestions align with user intent, while steerability enhances adaptability [15]. Revision-focused tools like Soylent [6], ABscribe [31], and RevTogether [38] treat AI as a second set of eyes, making the revision process more rewarding and fruitful [40]. SYNthia builds on these by allowing real-time refinement of word suggestions, maximizing user control.

2.3 The Good, the Bad, and the Ugly

AI in writing can increase writer productivity and confidence, aid in brainstorming, and improve clarity [22, 27]. However, concerns about "AI-giarism" (AI-driven plagiarism) and loss of ownership also emerge [9, 12, 18]. The increasing difficulty of distinguishing between AI-generated and authentic human writing creates a loss of ownership for the writers [12, 18], and distrust from readers [18].

2.4 Effective Error Correction

Implementing efficient error correction has long been a subject of discussion [8]. With algorithms such as handwriting, speech, and text recognition models increasing in popularity [3, 23, 25, 36, 37, 39], their susceptibility to error was quickly uncovered [37]. While such recognition models could only achieve a certain threshold of accuracy, user satisfaction could still be achieved through the process of error correction, whether it be automatic or manual. However, it is important to note that most error correction papers concern themselves with the scope of error correction in transcription/recognition contexts, and not necessarily in the context of writing or word choice. Although our project does not focus on the fusion of text and voice inputs, we incorporate additionally provided textual context by the user as a form of error correction, a process described by Davis as human-computer co-creativity [11].

3 SYNthia

SYNthia, derived from "synonyms," is an interactive AI-powered thesaurus. Unlike traditional thesauruses, SYNthia's clean interface minimizes context switching with a side-by-side writing interface. In addition, SYNthia allows writers to completely control the number of suggestions (capped at 10 to prevent cognitive overload) and provide context to refine results.

3.1 Design Goals

SYNthia attempts to fulfill three primary objectives: coherence to context, steerability of suggestions, and divergent outcomes, aligning with prior research [11, 15]. Coherence to context ensures relevant suggestions [15], while steerability allows users to refine outputs dynamically [11]. The ability to give context directly influences SYNthia's suggestions, following a fixed LLM prompt (Appendix B), addressing the lack of efficient steerability in prior tools. To enhance divergence, SYNthia lets users adjust the number of suggestions, supporting variation in writers' work. A full system architecture description is available in Appendix A.

4 Methodology

We chose to conduct interviews to gain rich, first-hand insights into participants' writing processes, tool usage, and perceptions, information that would be difficult to capture through surveys alone. We interviewed 19 participants (12 novice, 7 expert writers) from US colleges and Prolific, choices motivated by funding and familiarity with Prolific's recruiting pipeline. We defined experts as writers with published works and/or writing degrees (e.g., MFA) in English or fiction writing; others were labelled as novices. Participants completed a screening survey (Appendix B) and received $25 for their interview. Each 45-minute Zoom session included pre/post-interview questions and a 15-minute SYNthia-assisted writing task. Interviews were recorded with consent and transcribed via Zoom. Quantitative analysis was done on metrics like number of suggestions requested, number of suggestions accepted, and time spent writing. Qualitative analysis (in particular, thematic analysis) was done on the pre- and post-interview responses. Full details are in Appendix B.

5 Results

5.1 Quantitative Overview

On average, participants requested nine synonyms per session, added context for two words, and accepted 43% of SYNthia's suggestions. We analyzed backend data on word prompts, context use, acceptance rates, and time spent writing and using SYNthia. Independent t-tests for word prompts and time spent showed no significant differences between novice and expert writers. However, their interactions with SYNthia and the influence of context on their cognitive processes (as demonstrated by participant self-reports in the interviews) varied.

5.2 Context Feature

SYNthia's key feature is user feedback through context input. A representative taxonomy of the types of contexts can be found in Table 1, with examples of each context type. These prompts given by participants were largely dependent on the content / direction of their writing, as well as the specific types of words they were looking for in that moment. Four of seven experts said suggestions didn't affect their writing, and three found the context feature unhelpful in any capacity. In contrast, only three of twelve novices were unaffected by suggestions, and four found the context feature unhelpful. This section further breaks down the utility of the context-providing feature, as well as its cognitive impacts on the writers. The following subsections provide a breakdown of common themes we found across interviews after performing a detailed thematic analysis [7].

5.2.1 Utility of Context. Participants were motivated to use SYNthia's context feature due to its initially **unsatisfactory results**, as they needed "words that fit a specific context" (P13). All but two participants used the context feature, particularly when they **believed SYNthia could give a satisfactory suggestion with a slight nudge from them**. P5 described a step-by-step process: starting with "sad," refining the suggestions, and eventually finding "melancholy."

Participants provided context in four key scenarios. Sometimes, SYNthia suggested a **wrong part of speech**, as seen when P1 specified "esoteric noun" for the word "place." Others encountered words with the **wrong tense**, like P14 refining "woke" to "awakened" by specifying that they wanted a "past tense verb." **Words with multiple meanings** also posed challenges. For example, P18 wanted to use the word "lead" as the heavy metal element, rather as the act of guiding/leading. Finally, context was crucial for achieving a **particular feeling or tone**, as P15 refined "random" to "arbitrary" by specifying "more old fashioned" (see Table 1).

5.2.2 Cognitive Impacts of Context. While most participants found the context feature helpful, a handful of participants found the act of giving context **energy-intensive and sometimes confusing**. Some struggled to "describe the context" (P14) or doubted if "the context would make sense to the system" (P10), dissuading them from fully engaging with that feature.

Despite this, giving context and seeing suggestions **triggered critical thinking**, refining plot and word choice. P6 noted that refined suggestions inspired new story directions, while others found that rejecting suggestions still shaped their writing (P11, P12). Suggestions "challenged [their] thinking" (P16), encouraging participants to "write with their instinct" (P3).

Even when context led to unsatisfactory results, suggestions **validated original word choices**, boosting confidence. For writers who "second-guess themselves" (P15), seeing weaker alternatives reassured them their initial choice "was probably right" (P7). P16 explained, "Seeing more aligned words that weren't quite right made me feel more confident in my original choice." SYNthia, even when imperfect, reinforced user instincts.

6 Discussion

6.1 How Writing Skills Modulate SYNthia's Impact on Cognition

Overall, our study suggested that SYNthia's outputs had less impact on the thought process of expert writers, and more impact on the thought process of novice writers. This difference may stem from

Context Type	Writing Phrase	Context Given	Word Chosen
Part of speech	Across the street is a dense forest. It looks dark, **foreboding** .	Adjective	Ominous
Word tense	I **woke** with a jolt.	I'm using it as a past tense verb	Awakened
Pop culture reference	Where am I? How did I get here? Am I in **SquidGame** ?	Shows similar to the Netflix show Squid Game	Money Heist
Object description	He threw off the **ratty** sheets and slid his feet into disintegrating flip-flops.	To describe sheets	Threadbare
Specific tone	Maybe if I used the GPS then I would have found someone to help me instead of just calling a **random** number.	More old fashioned	Arbitrary
Words with multiple meanings	My hands felt heavy like **lead** .	Weight heavy, not the act of leading	N/A

Table 1: Taxonomy of contexts given by participants

how each group perceives SYNthia's role—novices may see AI as authoritative, whereas experts may treat it as a "junior collaborator" [20]. This power dynamic may lead novices to over rely on AI, reducing stylistic confidence [35], whereas experts prioritize maintaining their own authentic creative processes [19]. In SYNthia's design, we aimed to protect writer cognition [34]. Participants who fully engaged with SYNthia deepened their cognitive involvement, while those who "got lazy" (P6) saw little impact. User interaction, more than the tool itself, shaped the outcome.

6.2 Trade-offs Between Traditional and AI-powered Thesauruses

SYNthia offers several advantages over traditional thesauruses, but also presents trade-offs. DG1 highlights that SYNthia provides contextually relevant and coherent suggestions, with 14 out of 19 participants expressing satisfaction with how the tool aligned with the context they provide. DG2 emphasizes SYNthia's steerability, enabling it to adapt outputs based on user input, which fosters co-creativity while maintaining user agency. However, DG3 reveals a trade-off: while SYNthia generates diverse, contextually relevant suggestions, some participants felt that traditional thesauruses offer a broader range, albeit with more irrelevant options. SYNthia introduces a trade-off between quantity and relevance—too many suggestions can overwhelm users, while too few may limit exploration. Future work could focus on balancing the relevance and quantity of synonym suggestions while optimizing LLM prompts for greater diversity.

7 Limitations and Future Works

The fixed 15-minute prompt prioritized storytelling/drafting over editing, with many participants feeling dissatisfied with their rushed writing. In addition, opinions on automation varied. While most found the manual copy-pasting mechanism intuitive, some preferred automatic word replacement. Others wanted SYNthia to read surrounding text for context (P1). Future research could explore how these features affect usability and writer agency.

7.1 Additional Interface Features

SYNthia has improved upon its 2023 prototype, but further refinements remain:

(1) **History of prior prompts:** Users found that past suggestions "gave them ideas for later" (P17). A collapsible history of prompts, context, and suggestions could improve usability.
(2) **More breadth of suggestions:** Some participants disliked the shorter list, noting that thesauruses work by offering many options (P1). A "show more" button could offer more breadth if the user wants.
(3) **Phrases:** Unlike a traditional online thesaurus, SYNthia currently focuses on single words. Allowing phrase-based prompts via brackets or parentheses could address this limitation.

8 Conclusion

As AI becoming increasing prominent in creative fields, concerns about its threat to human agency continue to grow, particularly among writers. In addition, AI hallucinations and inaccurate outputs further push writers from considering them as potentially helpful tools. These concerns are addressed by SYNthia, created to show writers the capabilities of AI to assist them, and show developers how to create useful tools that do not pose a threat for writers. Our user study suggests that word-level AI suggestions do not compromise writer ownership and can be welcomed. Systems like SYNthia may foster trust writer trust in AI, making progress towards a world where the two fields live more harmoniously without taking away from either.

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A Full System Architecture

A.1 System Architecture

The interface starts with an empty text box on the left side and no suggestion boxes on the right side. The user can type in the text box and trigger SYNthia by appending an @ symbol to the end of the word for which they want alternative suggestions.

The word associated with the @ symbol is then sent through a GPT-4 prompt, which outputs an initial list of synonyms in a list format. That list is then parsed and displayed on the HTML page. Users also have the ability to provide additional context to update the list of synonyms within the suggestion box. After pressing the "enter" key or the "submit" button, the user provided context is appended to the original GPT-4 prompt that regenerates a new list of synonyms. Ultimately, users can click on one of the synonyms (blue box) to copy that word to their clipboard, which indicates to the backend that the user has "chosen" that particular SYNthia suggestion. In addition to being shown on the user-facing interface, all words that trigger SYNthia, corresponding context, and chosen suggestions are added to a JSON file for further analysis.

B Full User Study Protocol

B.1 Prompts

B.1.1 Writing Task Prompt. You woke up in a motel room with a strange item in your hand. What do you do? Where do you go? Describe the world around you and your actions. Choose between a) a cell phone that can only call one specific number, b) a GPS to a mystery location 1 mile away, or c) a sticky note with a list of seemingly random items, written in dried blood.

B.1.2 LLM (GPT 4) Prompt. From now on, give me **numSuggestions** distinct synonyms for the word **word**. You MUST format your response as an array, for example: ["word1", "word2", "word3", "word4"]. Do not include any other information in your response. If you cannot come up with **numSuggestions**, provide as many as you can. All of my future messages will provide extra context for the word, you should incorporate them into your suggestions. The words you respond with should ALWAYS be synonyms for **word** and match the additional context.

numSuggestions = Number of suggestions as selected in slider
word = Word for which synonyms are to be generated

B.2 Participant Recruitment Survey Questions - Required

(1) First Name, Last Name (optional)
(2) Email
(3) Age (optional)
(4) Gender (optional)
(5) If you are a student, what do you study?
(6) What is your editing style? Do you edit as you write, or do you edit after drafting the full piece?
(7) How often do you find yourself thinking about alternative words during writing?
(8) Do you use any writing aids while you write (thesaurus / generative AI / Grammarly)? Please list them below.

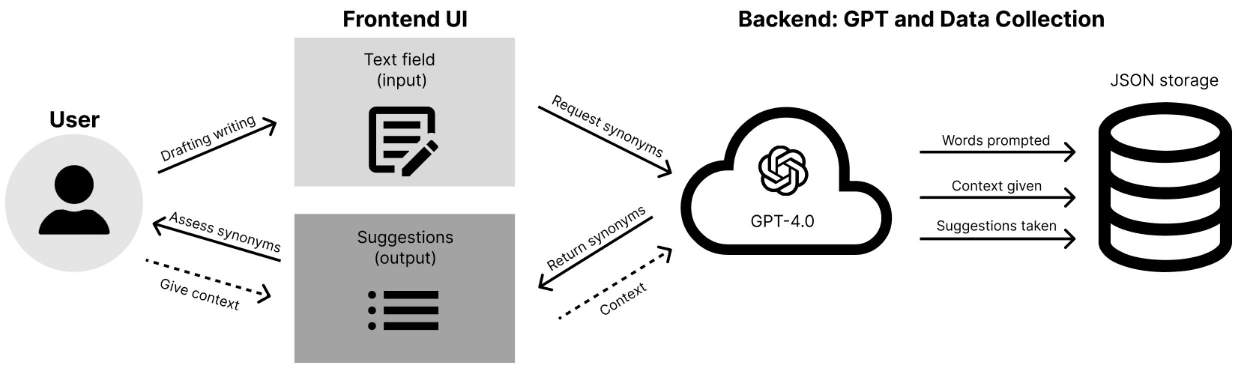

Figure 2: SYNthia system architecture
Alt Text: flowchart detailing the interactions between a user, the frontend, and OpenAI powered backend

(9) On a scale of 1 (worst) to 10 (best), how would you rank your creative writing skills? To help guide you: a 5 is equivalent to the skills of someone who is coming out of a MFA program (masters degree in writing).
(10) How many years of creative writing experience do you have?
(11) Have you received formal education in creative writing (e.g. degrees, classes, etc.)? If so, please list them here.
(12) How often do you use AI?
  (a) If your answer was not "Never" to the previous question, please list your prior AI experiences.
(13) How comfortable / confident do you feel with using generative AI and prompting them to get satisfactory results (1 - least comfortable; 10 - most comfortable)?

B.3 Pre-Interview Questions - Required

(1) How many hours per week do you write on average?
(2) Do you have any published works and/or manuscripts?
(3) What are your opinions on using AI in creative writing?

B.4 Play Test with SYNthia

Refer to the prompt in the *Writing Task Prompt* section. During the 15 minute writing process, participants will think out loud if they want, which will be recorded and transcribed. In addition, some quantitative measurements will also be recorded, including but not limited to:

(1) How often user prompted SYNthia
(2) Words SYNthia is prompted on
(3) Additional context given
(4) Number of reprompts
(5) Whether user ultimately chose a suggestion by SYNthia or not
(6) How smoothly users transition between using SYNthia and writing
(7) After prompting SYNthia, how long until they start writing again?
(8) Time spent writing vs. interacting with SYNthia vs. thinking
(9) Time spent engaging with SYNthia happens when:
(10) User has an @ symbol present in the text box
(11) User gives context to the suggestion box at any point, even if there is no active @ symbol present, and they are just going back to a previous suggestion

B.5 Post-Interview Questions - Required

(1) How did you decide when to give extra context?
(2) Did you think giving context changed the suggestions significantly?
(3) Did you feel like giving context to SYNthia was similar to
(4) prompting a generative AI (e.g. ChatGPT)?
(5) Were there any instances where you didn't take the specific word suggestion, but drew inspiration from the given suggestions?
  (a) Were you satisfied with the word suggestions given by SYNthia?
(6) How satisfied are you with the final product, from a scale of 1 to 10, with 10 being most satisfied (no notes) and 1 being least satisfied? Explain.
  (a) Do you think the use of SYNthia improved the quality of your writing, compared to if you were to write without any external help?
(7) Do you feel like this is your piece of writing, even though you used AI assistance? Rate your ownership level from 1 to 10, with 10 being "I feel like this writing was purely produced by me", and 1 being "I feel like this writing is completely AI generated".